Precalculus covers the introduction and evaluation of trigonometric functions, trigonometric identities, complex numbers, exponential form of complex numbers, De Moivre's Theorem, geometric representation of complex numbers, roots of unity, applications of complex numbers to geometry, two-dimensional and three-dimensional vectors and matrices, determinants, dot and cross product, applications of vectors and matrices to geometry.

This course is specifically designed for high-performing students and draws material from many programs for top high school students in the country. Our philosophy is that students develop more by learning to solve problems they haven't seen before, as opposed to offering repeated drills that students can memorize their way through. In this way, our classes are structured much more like courses at top-tier colleges.

Book: *Precalculus*, by Richard Rusczyk

Time Commitment: This 36-week course includes 105 minutes in-class and 1-3 hours of homework each week, corresponding to a full year course.

Content:

Week	Topic
1	Algebra & Geometry Review
2	Functions Review
3	Introduction to Trigonometry
4	Graphs of Trig Functions
5	Inverse Trig Functions
6	Angle Addition Formulas
7	Double & Half Angle Formulas
8	Trigonometric Equations
9	Parametric Equations
10	Polar Coordinates
11	Polar Graphing
12	**Midterm 1**
13	Laws of Sines and Cosines
14	Complex Numbers
15	Polar Form of Complex Numbers
16	Exponential Form of Complex Numbers
17	Roots of Unity
18	Complex Number Geometry I

Week	Topic
19	Complex Number Geometry II
20	Vectors in 2D
21	More Vectors in 2D
22	Projections in 2D
23	Vector Geometry
24	**Midterm 2**
25	Matrices in 2D
26	Matrices as Transformations in 2D
27	Inverse Matrices in 2D
28	Determinants in 2D
29	3D Coordinates
30	Vectors in 3D
31	Cross Product
32	Planes in 3D
33	Projections in 3D
34	Matrices in 3D
35	Determinants in 3D
36	**Final Exam**

Common Core State Standards:

Domain	Subdomain	Standards
Number and Quantity	The Complex Number System	4, 5, 6, 7, 8, 9
	Vector & Matrix Quantities	1, 2, 3, 4abc, 5ab, 6, 7, 8, 9, 10, 11, 12
Algebra	Seeing Structure in Expressions	3a
	Arithmetic with Polynomial & Rational Expressions	1, 2, 3
	Reasoning with Equations and Inequalities	8, 9
Functions	Interpreting Functions	5, 7e
	Building Functions	1c, 3, 4abcd
	Trigonometric Functions	1, 2, 3, 4, 5, 6, 7, 8, 9
Geometry	Congruence	2, 3, 4, 5, 6
	Similarity, Right Triangles, & Trigonometry	7, 8, 9, 10, 11
	Expressing Geometric Properties with Equations	1

Sample Problems:

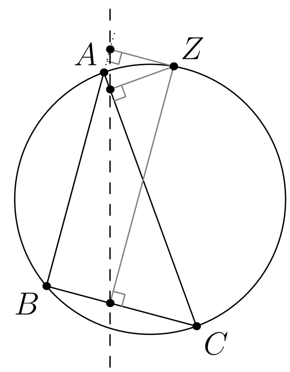
►

Let point D be on side $\overline{BC}$, and let $d = AD, m = BD$, and $n = CD$. Stewart's Theorem states that

$$a(d^2 + mn) = mb^2 + nc^2.$$

Using trigonometry, prove Stewart's Theorem.

► Simson Line: Let Z be a point on the circumcircle of $\triangle ABC$ besides A, B, and C. Use complex numbers to show that the feet of the perpendiculars from Z to the lines containing the sides of $\triangle ABC$ are collinear.



► Find a matrix $\mathbf{N}$ such that $\mathbf{N}\mathbf{v}$ is the projection of $\mathbf{v}$ onto the graph of $x + y + z = 0$.

